Music-Inspired Optimization Algorithm

Harmony Search

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What is Optimization?

- Procedure to make a system or design as effective, especially the mathematical techniques involved. (or Meta-Heuristics)
- Finding Best Solution
  - Minimal Cost (Design)
  - Minimal Error (Parameter Calibration)
  - Maximal Profit (Management)
  - Maximal Utility (Economics)

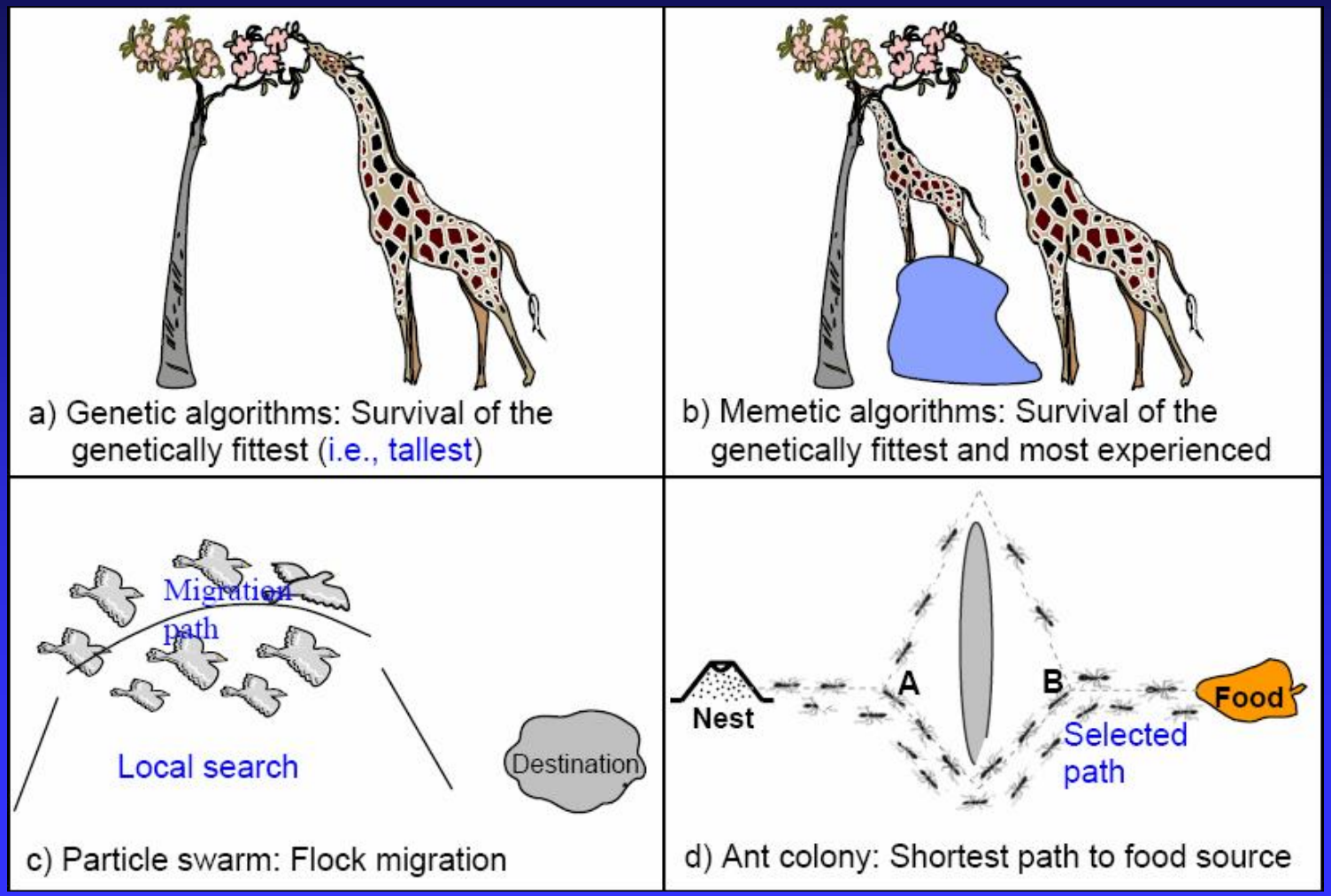
Types of Optimization Algorithms

- Mathematical Algorithms
  - Simplex (LP), BFGS (NLP), B&B (DP)
- Drawbacks of Mathematical Algorithms
  - LP: Too Ideal (All Linear Functions)
  - NLP: Difficult to Find Derivative, Feasible Initial Vector
  - DP: Exhaustive Enumeration
- Meta-Heuristic Algorithms
  - GA, SA, TS, ACO, …

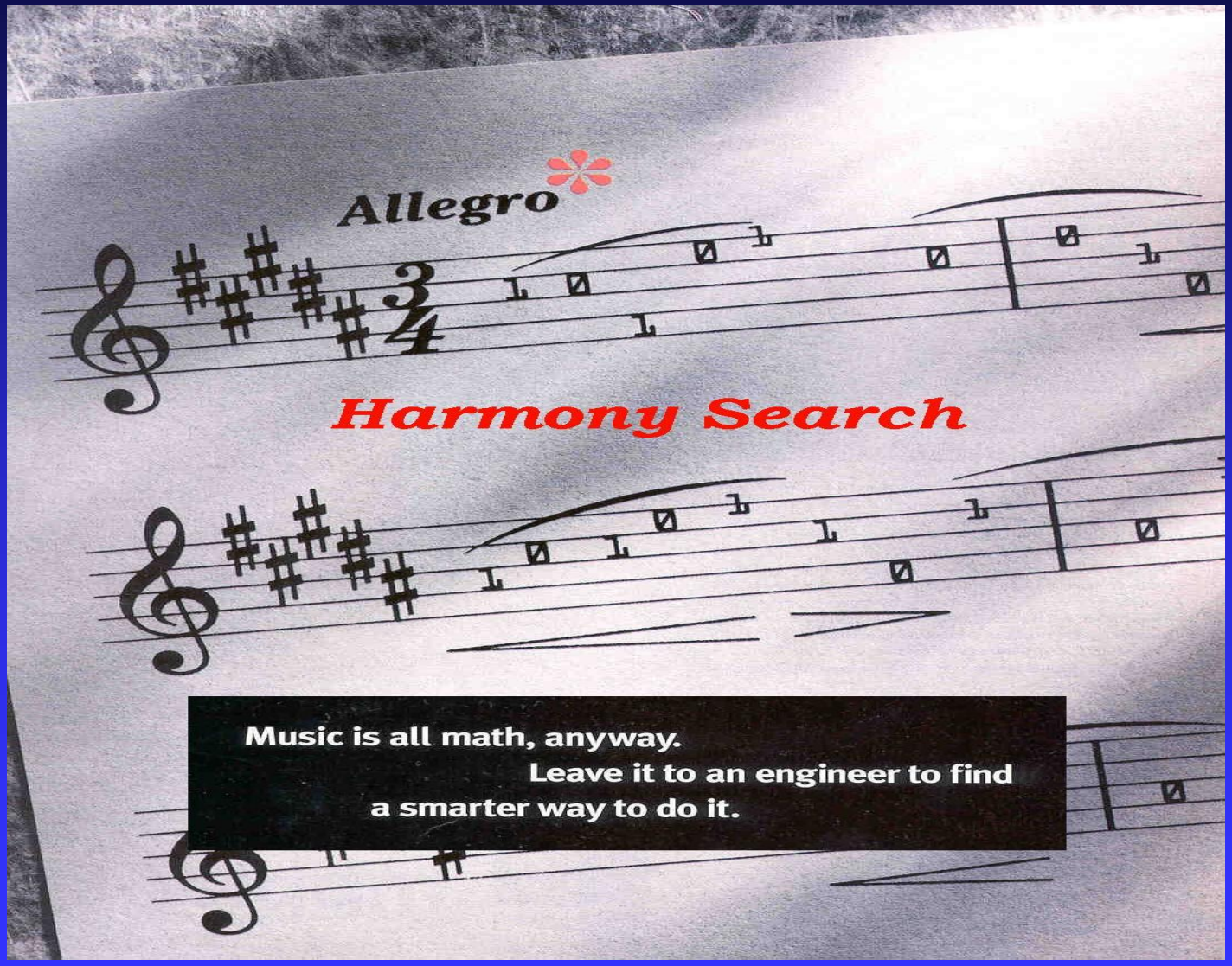
Existing Meta-Heuristic Algorithms

- Evolutionary Algorithm (Evolution)
- Simulated Annealing (Metal Annealing)
- Tabu Search (Animal's Brain)
- Ant Algorithm (Ant's Behavior)
- **Mimicking Natural or Behavioral Phenomena**

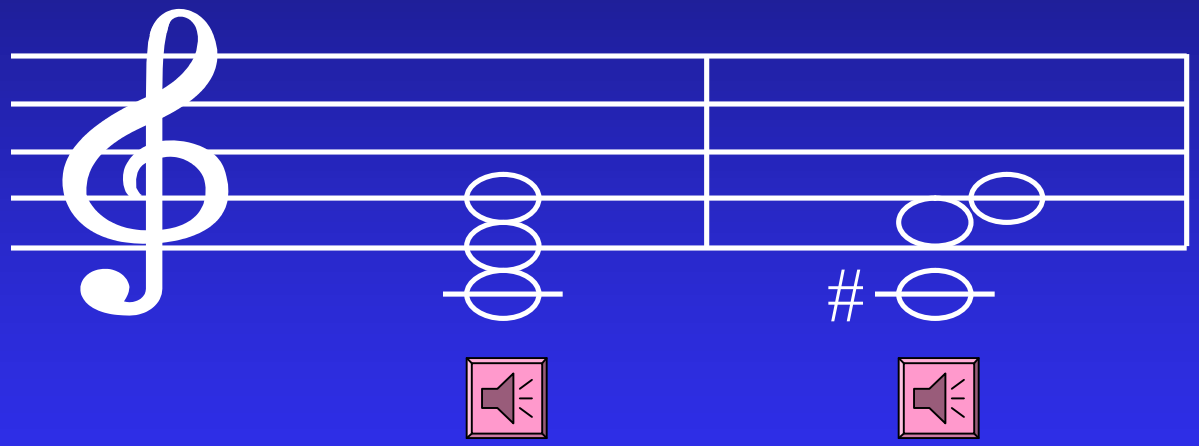
Existing Nature-Inspired Algorithms



Algorithm from Music Phenomenon



Good Harmony & Bad Harmony



An Algorithm which Keeps Better Harmonies!

Analogy

= Do = Mi = Sol

x_1 x_2 x_3

100mm 200mm 300mm **300mm 400mm 500mm** **500mm 600mm 700mm**

= 100mm = 300mm = 500mm

$f(100, 300, 500)$

Comparison Factors

COMPARISON FACTOR	OPTIMIZATION PROCESS	PERFORMANCE PROCESS
Best state	Global Optimum	Fantastic Harmony
Estimated by	Objective Function	Aesthetic Standard
Estimated with	Values of Variables	Pitches of Instruments
Process unit	Each Iteration	Each Practice

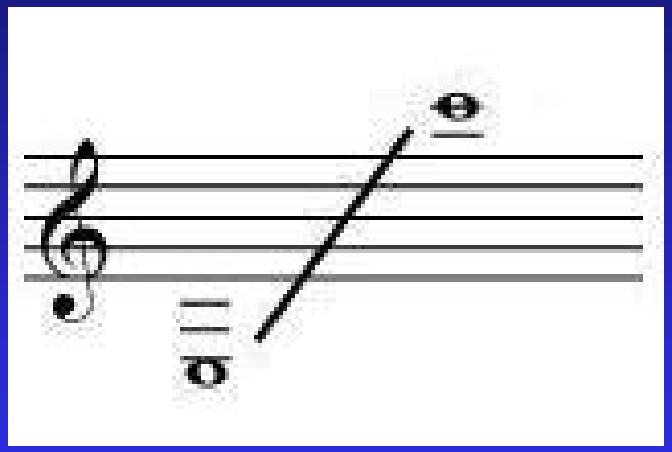
Procedures of Harmony Search

- Step 1. Improvise a new Harmony.
- Step 2. If the new Harmony is better, include it in Harmony Memory.
- Step 3. Repeat Step 1 and Step 2.

HS Operators

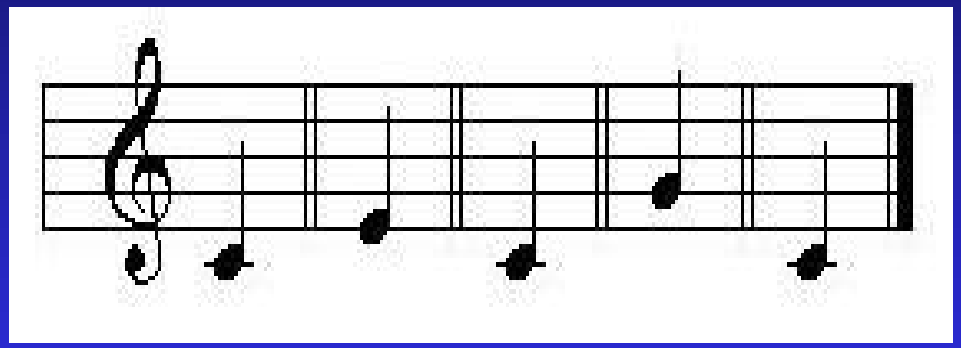
1. Random Playing
2. Memory Considering
3. Pitch Adjusting
4. Ensemble Considering
5. Dissonance Considering

Random Playing



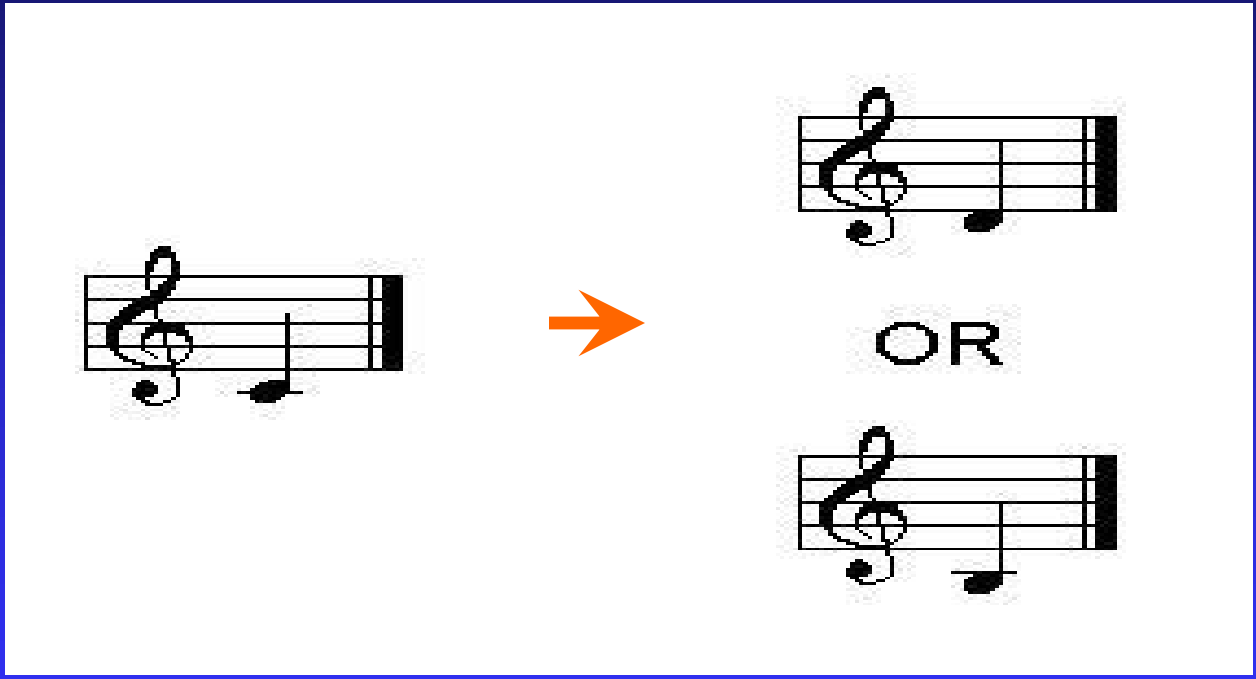
$x \in$ Playable Range = {E3, F3, G3, A3, B3, C4, D4, E4, F4, G4, A4, B4, C5, D6, E6, F6, G6, A6, B6, C7}

Memory Considering



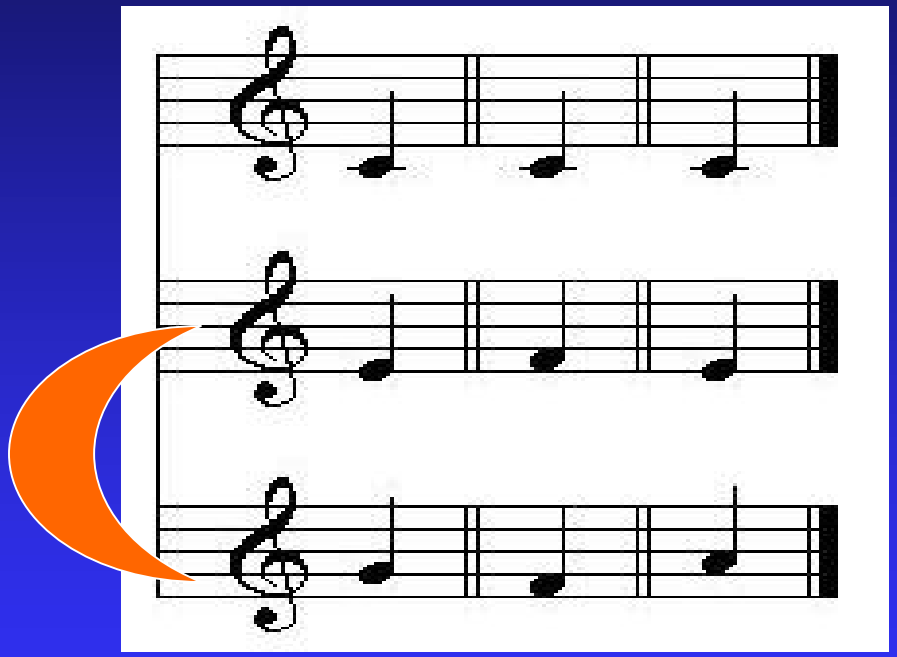
$x \in$ Preferred Note = {C4, E4, C4, G4, C4}

Pitch Adjusting



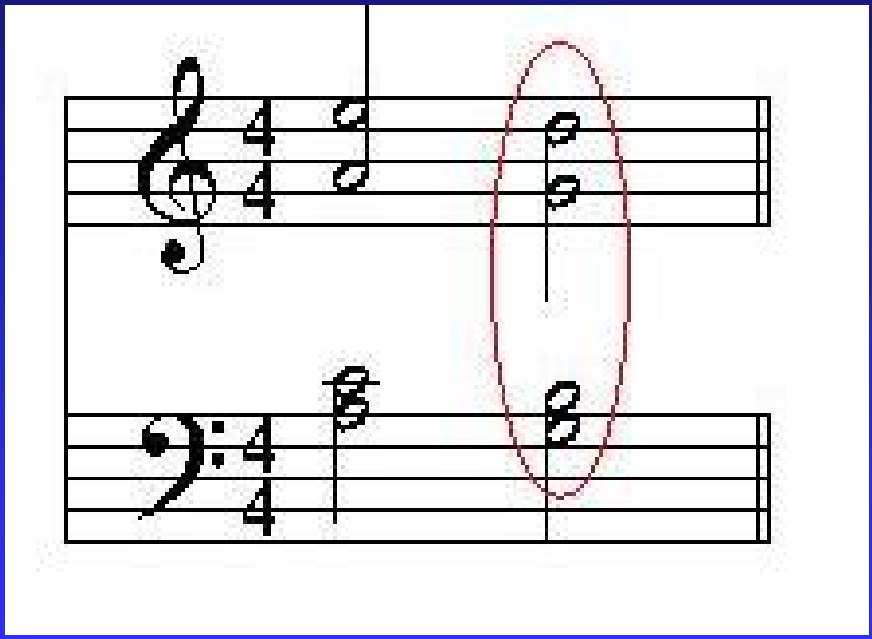
x+ *or x*-, *x* ∈ Preferred Note

Ensemble Considering



$$x_i \leftarrow f\left(x_j\right), \underset{j}{Max}\left\{Corr\left(x_i, x_j\right)\right\}$$

Dissonance Considering



$$Penalty = a\,(Violation\,Amount)^{b} + c$$

Example of Harmony Search

$$Min \quad f(\mathbf{x}) = (x_1 - 2)^2 + (x_2 - 3)^4 + (x_3 - 1)^2 + 3$$

Solution Vector = (2, 3, 1)

Initial Harmony Memory

$$Min \quad f(\mathbf{x}) = (x_1 - 2)^2 + (x_2 - 3)^4 + (x_3 - 1)^2 + 3$$

	X_1	X_2	X_3	F
Rank 1	2	2	1	4
Rank 2	1	3	4	13
Rank 3	5	3	3	16

Next Harmony Memory

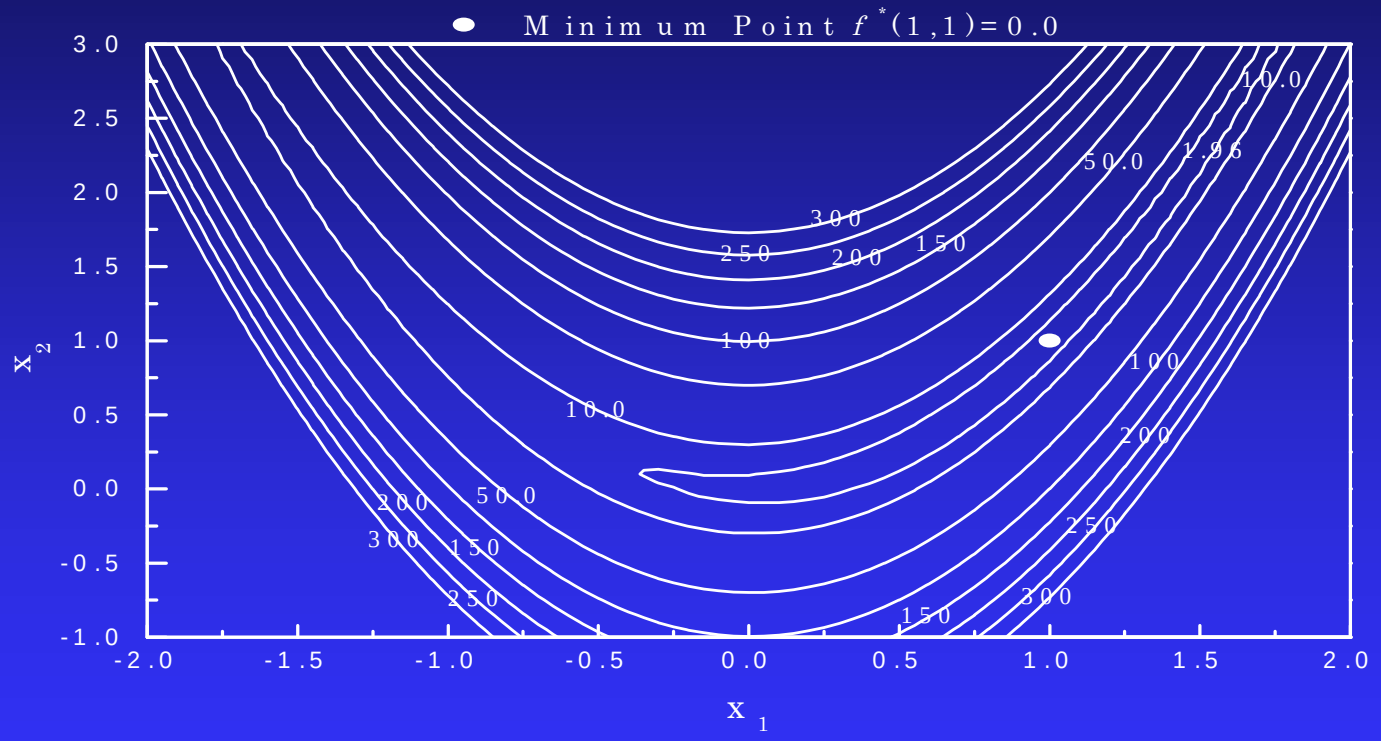
$$Min \quad f(\mathbf{x}) = (x_1 - 2)^2 + (x_2 - 3)^4 + (x_3 - 1)^2 + 3$$

	X_1	X_2	X_3	F
Rank 1	2	2	1	4
Rank 2	1	2	3	9
Rank 3	1	3	4	13

HS Applications for Benchmark Problems

Rosenbrock's Banana Function

$$\text{Min } f(\mathbf{X}) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$$

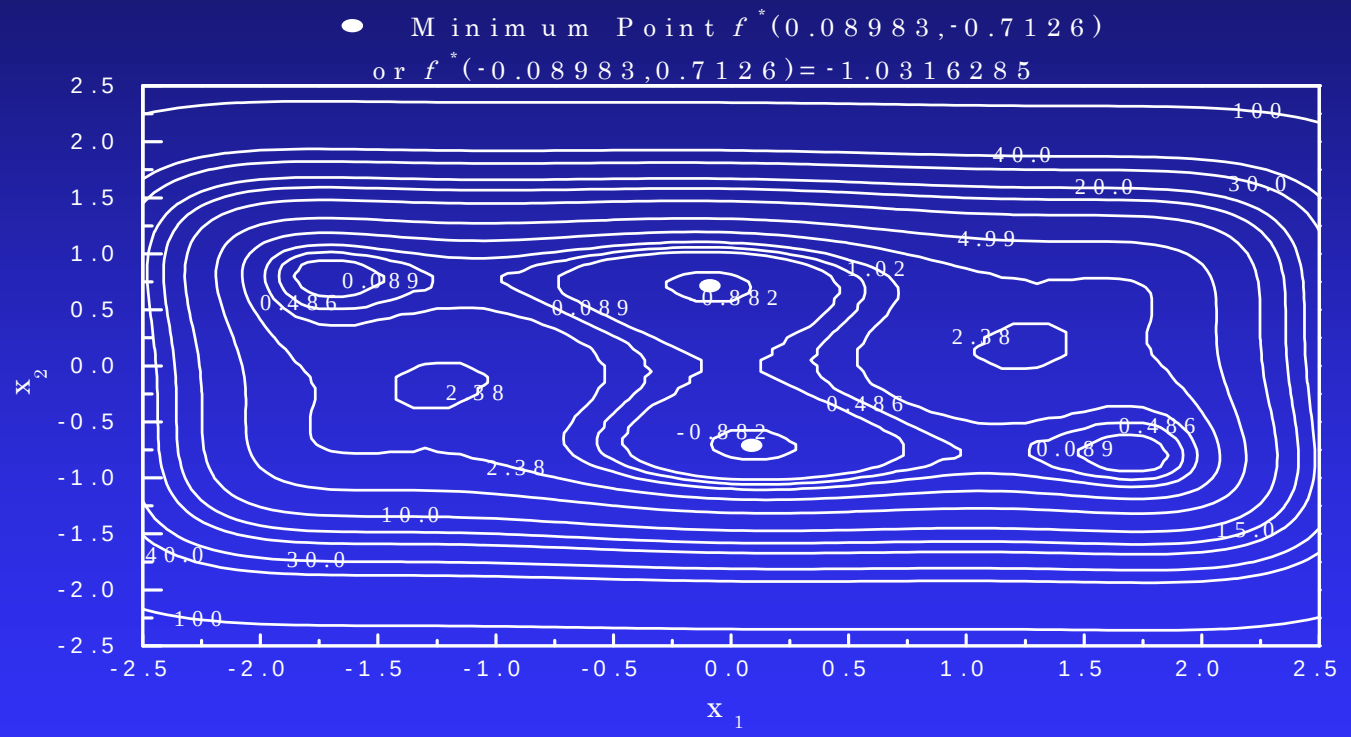


f*(1.000000, 1.000000) = 0.0 (Exact)

f (1.000062, 1.000137) = 0.00000002 (HS)

Six-Hump Camel Back Function

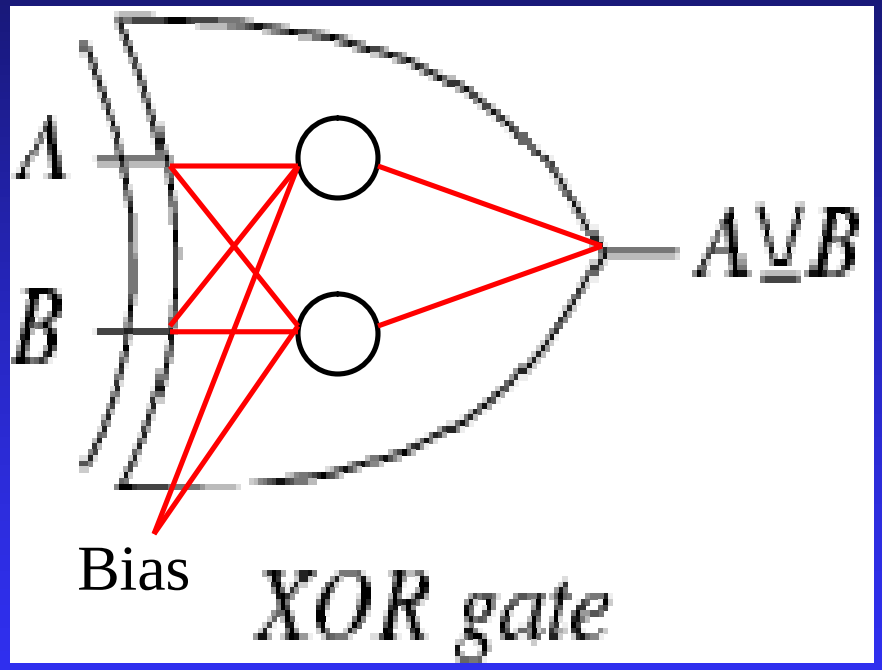
$$\text{Min}\, f(\mathbf{x}) = (4 - 2.1x_1^2 + \frac{1}{3}x_1^4)x_1^2 + x_1x_2 + (-4 + 4x_2^2)x_2^2$$



f*(-0.08983, 0.7126) = -1.0316285 (Exact)

f (-0.08975, 0.7127) = -1.0316285 (HS)

Artificial Neural Network - XOR



A	B	$A \veebar B$
T	T	F
T	F	T
F	T	T
F	F	F

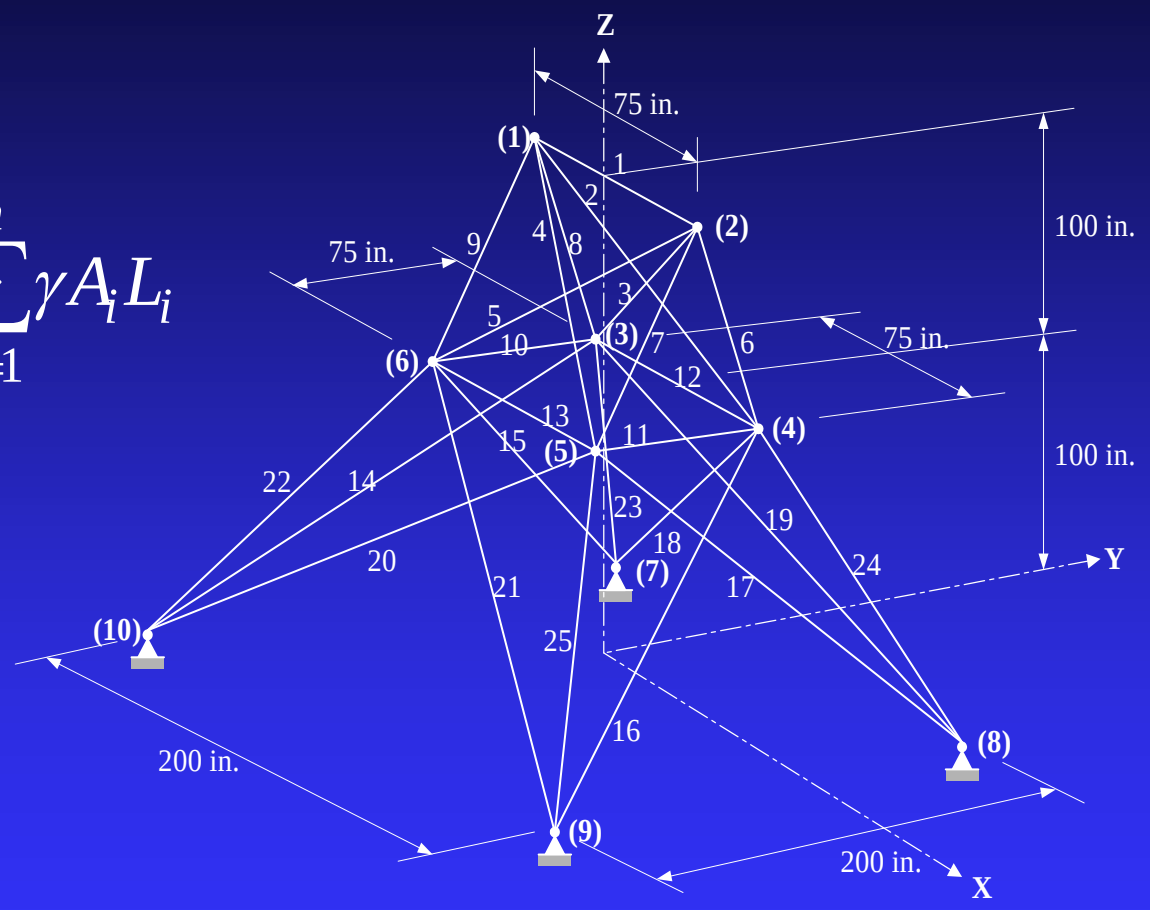
Sum of Errors in BP = 0.010

Sum of Errors in HS = 0.003

HS Applications for Real-World Problems

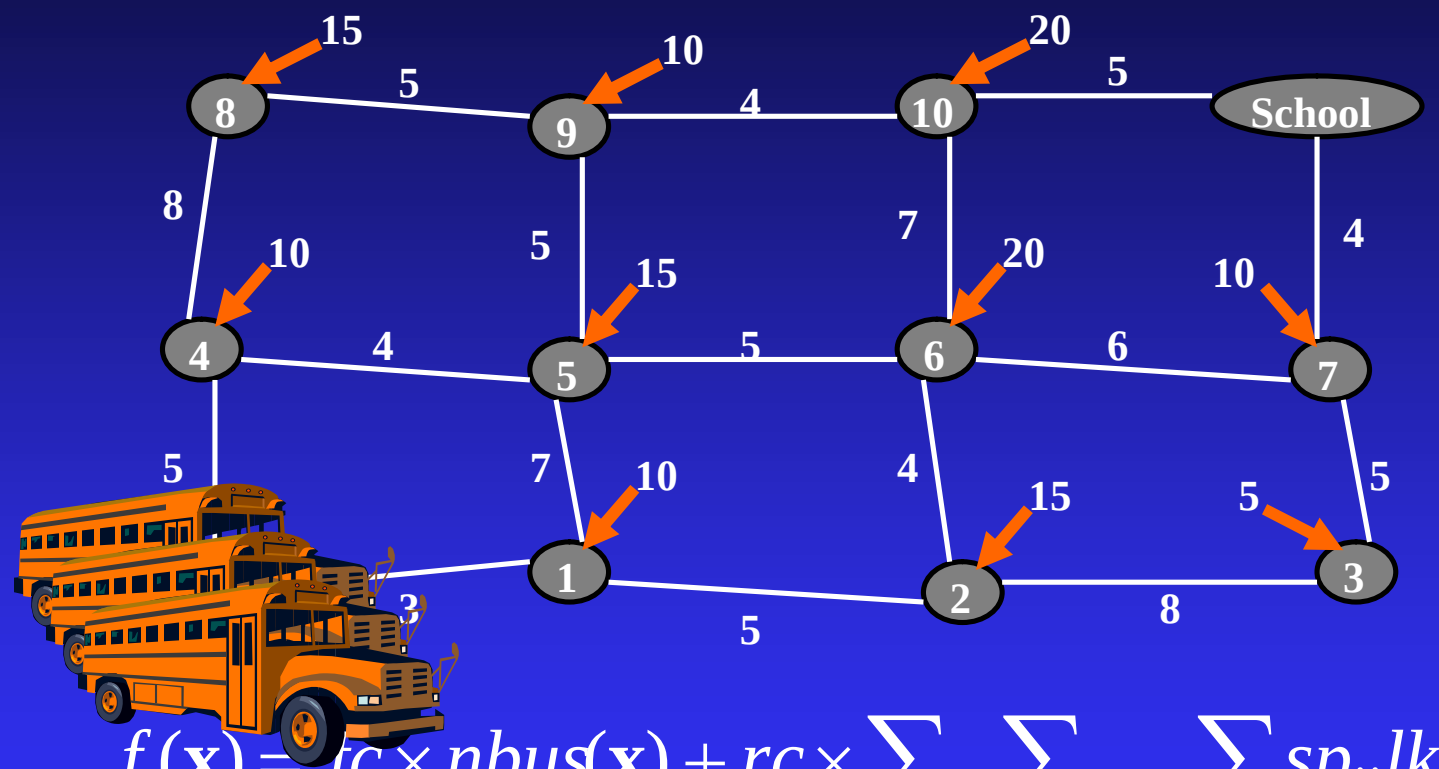
Truss Design – Minimal Weight of Structure

$$W(\mathbf{A}) = \sum_{i=1}^{n} \gamma A_i L_i$$



GA = 546.01, HS = 484.85

School Bus Routing Problem



$$f(\mathbf{x}) = fc \times nbus(\mathbf{x}) + rc \times \sum_{k} \sum_{i \in STDN} \sum_{j \in DNED} sp_{ij} lk_{ij}^{k}$$

GA = $409,597, HS = $399,870

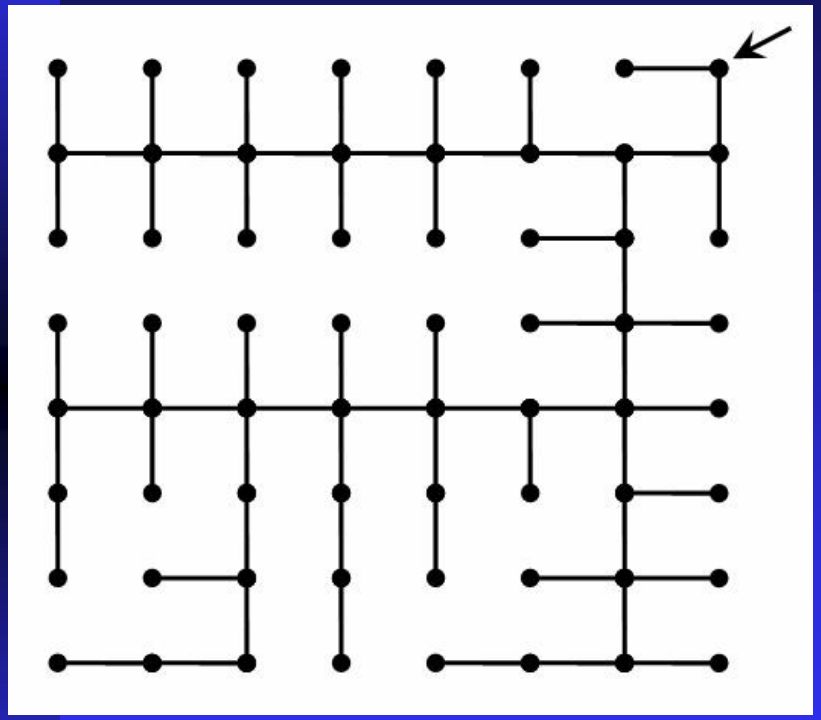
Generalized Orienteering Problem

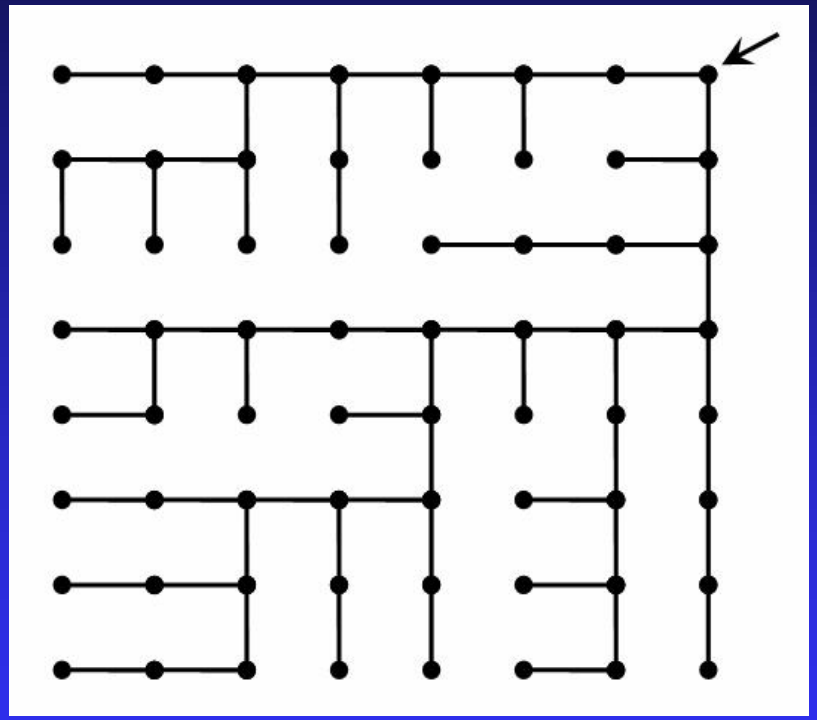
Multi-Objective Functions

1. Natural Beauty
2. Historical Significance
3. Cultural Attraction
4. Business Opportunity

	Case1	Case2	Case3	Case4	Case5
ANN	12.38	13.05	12.51	12.78	12.36
HS	12.38	13.08	12.56	12.78	12.40

Branched Network Layout



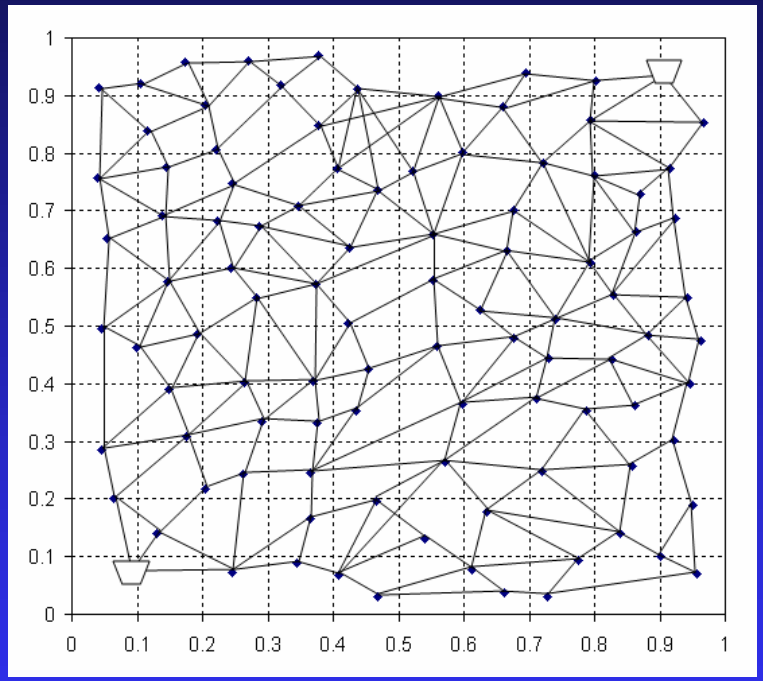


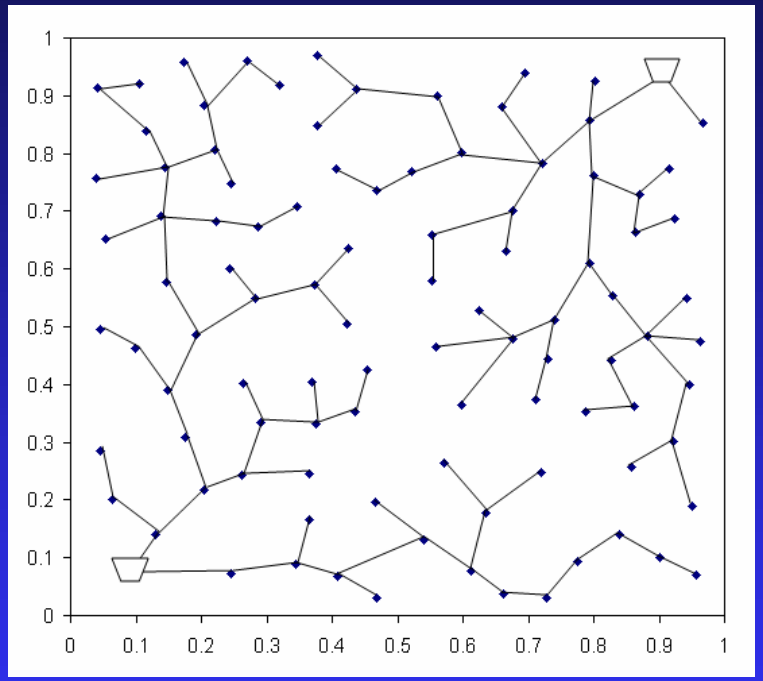
HS = 5062.8 (Global Optimum)

GA = 5218.0 (Near-Optimum)

Ant = 5095.2 (Near-Optimum)

Huge Branched Network Layout





100 Nodes, 231 Arcs

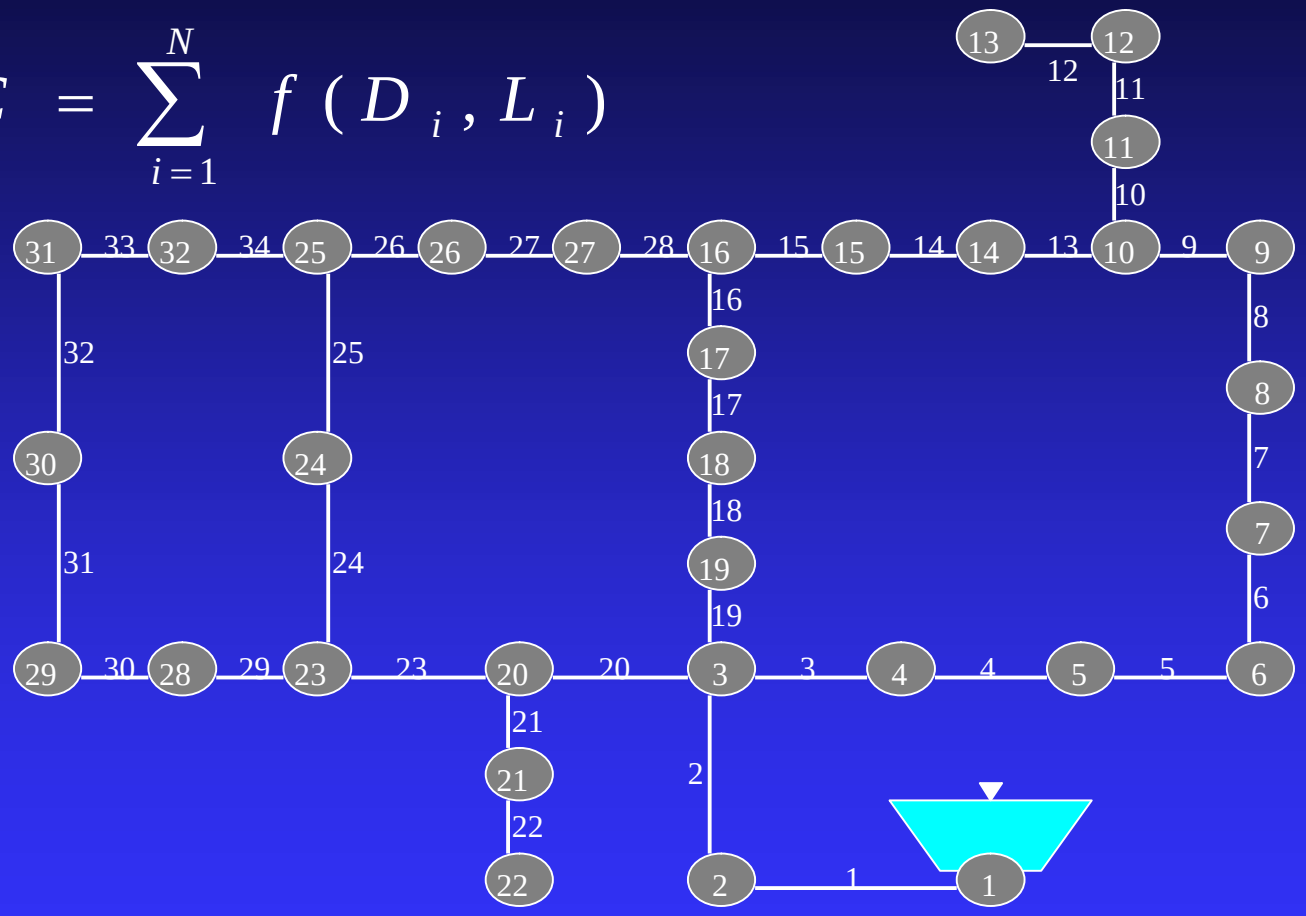
Ant = 31.13

EA = 28.22

HS = 28.06

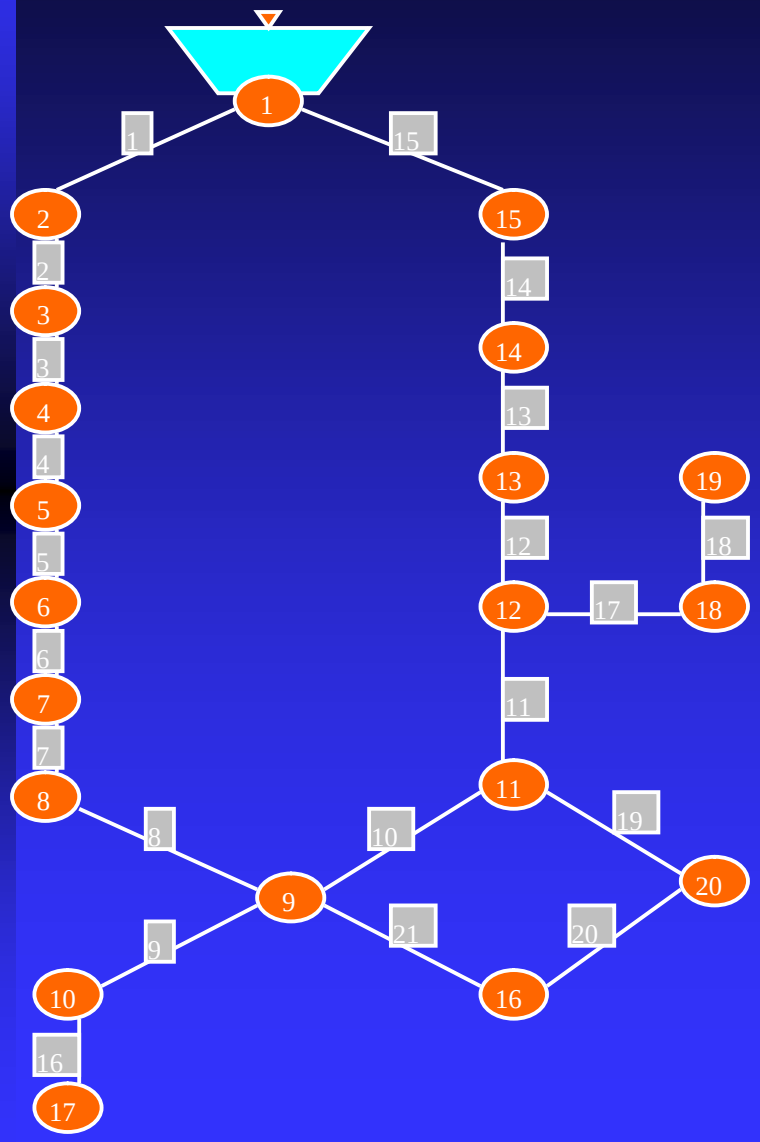
Water Distribution Network Design

$$C = \sum_{i=1}^{N} f(D_i, L_i)$$



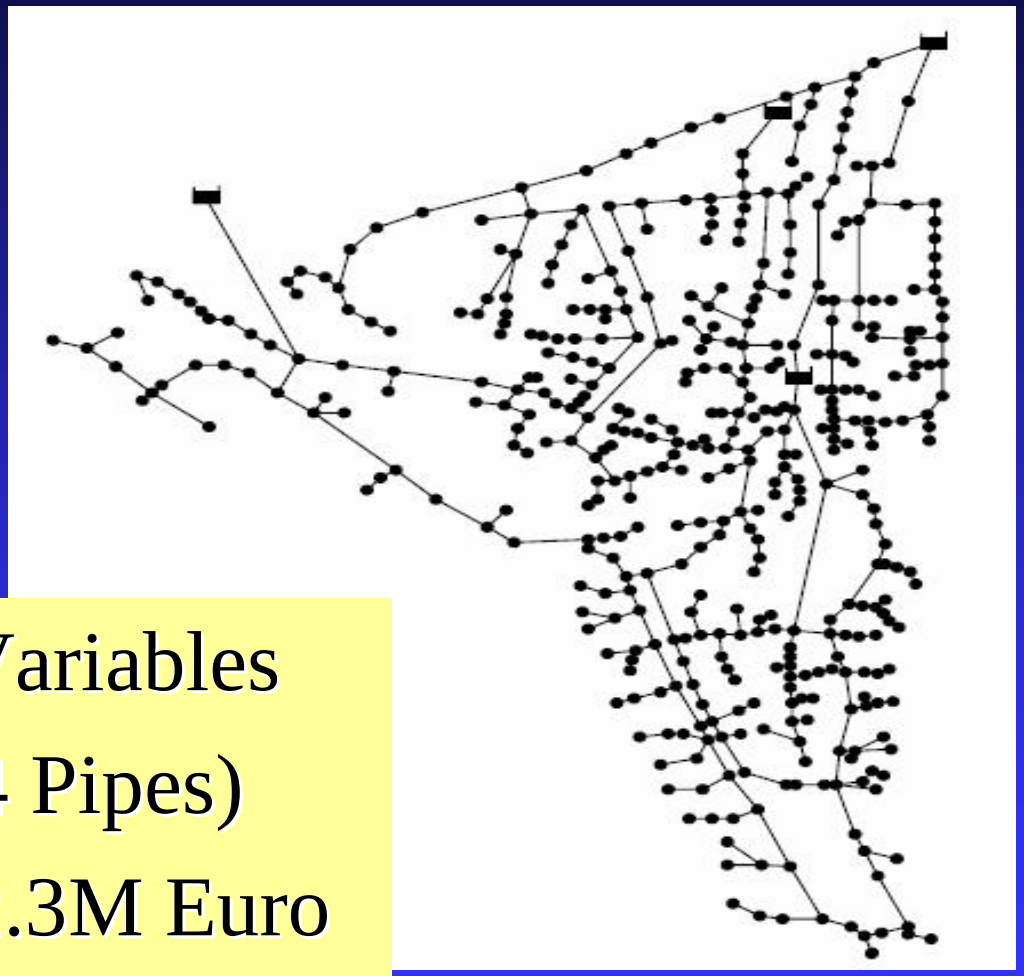
Mathematical = 6.320M, GA = 6.073M, HS = 6.056M

Water Distribution Expansion



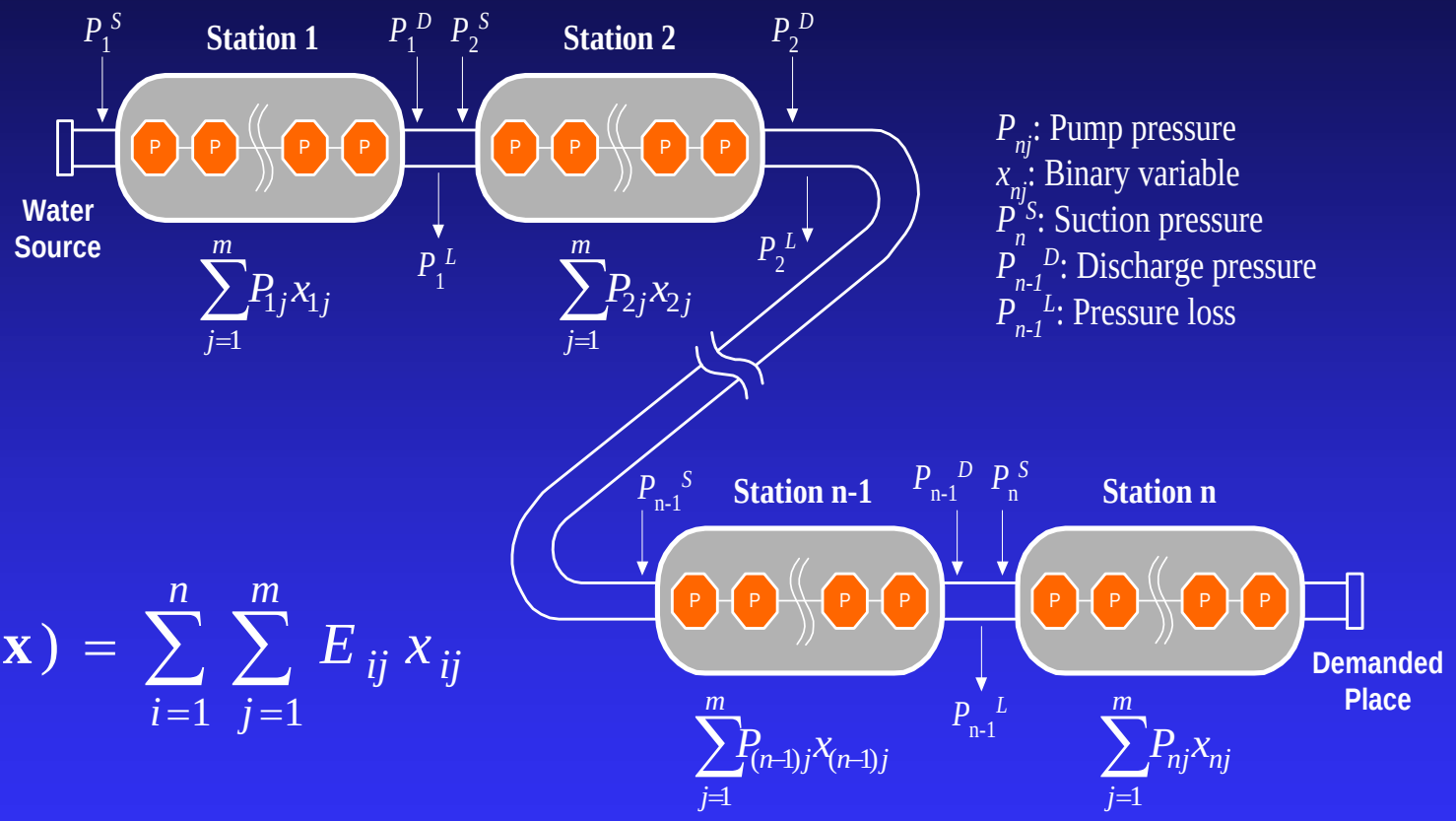
- MP: $78.09M
- GA: $38.64M (800,000)
- SA: $38.80M (Unknown)
- TS: $37.13M (Unknown)
- Ant: $38.64M (7,014)
- SFLA: $38.80M (21,569)
- CE: $38.64M (70,000)
- HS: $38.64M (3,373)
  - 5 times out of 20 runs

Huge Water Distribution Network



- Huge Variables
  - (454 Pipes)
- GA = 2.3M Euro
- HS = 1.9M Euro

Energy-Saving Pump Switching

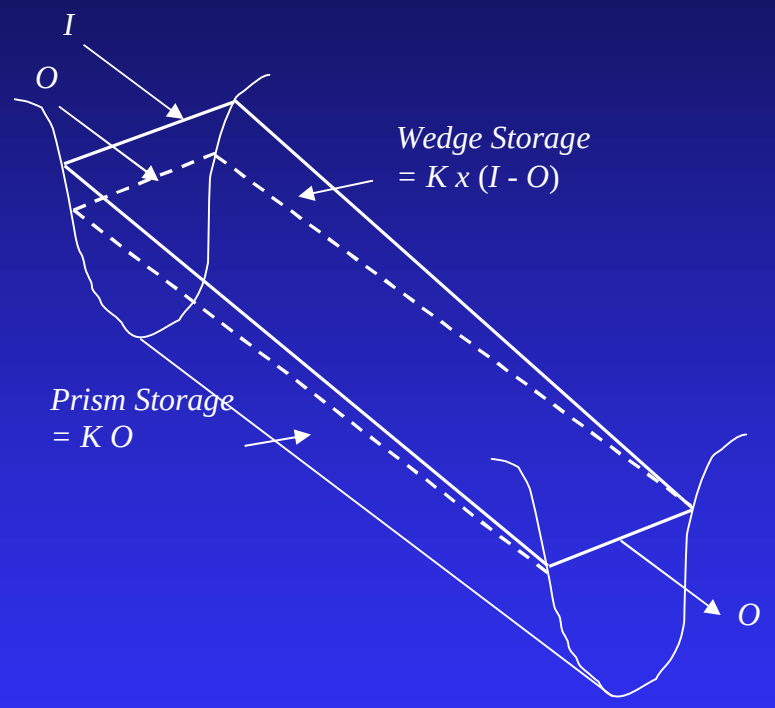


$$f(\mathbf{x}) = \sum_{i=1}^{n} \sum_{j=1}^{m} E_{ij} x_{ij}$$

GA = 11263.19, HS = 11164.47

B&B (IBM) = 11187.00, B&B (MS Excel) = 11181.37

Hydrologic Parameter Calibration

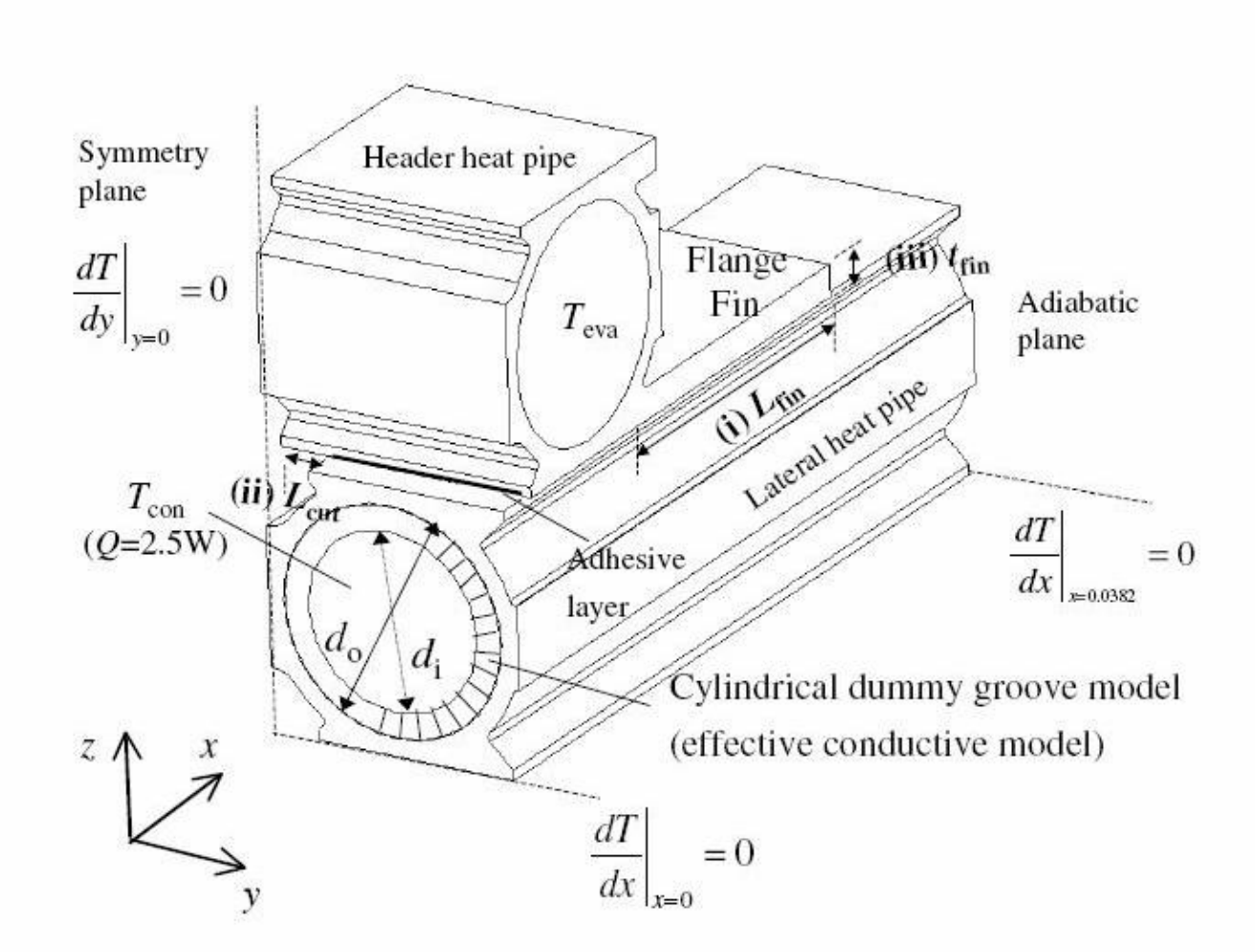


$$S_t = K\left[xI_t + (1-x)O_t\right]^m$$

$$\text{SSQ} = \sum_{t=1}^{n} \left(O_t - \hat{O}_t\right)^2$$

Mathematical = 143.60, GA = 38.23, HS = 36.78

Satellite Heat Pipe Design



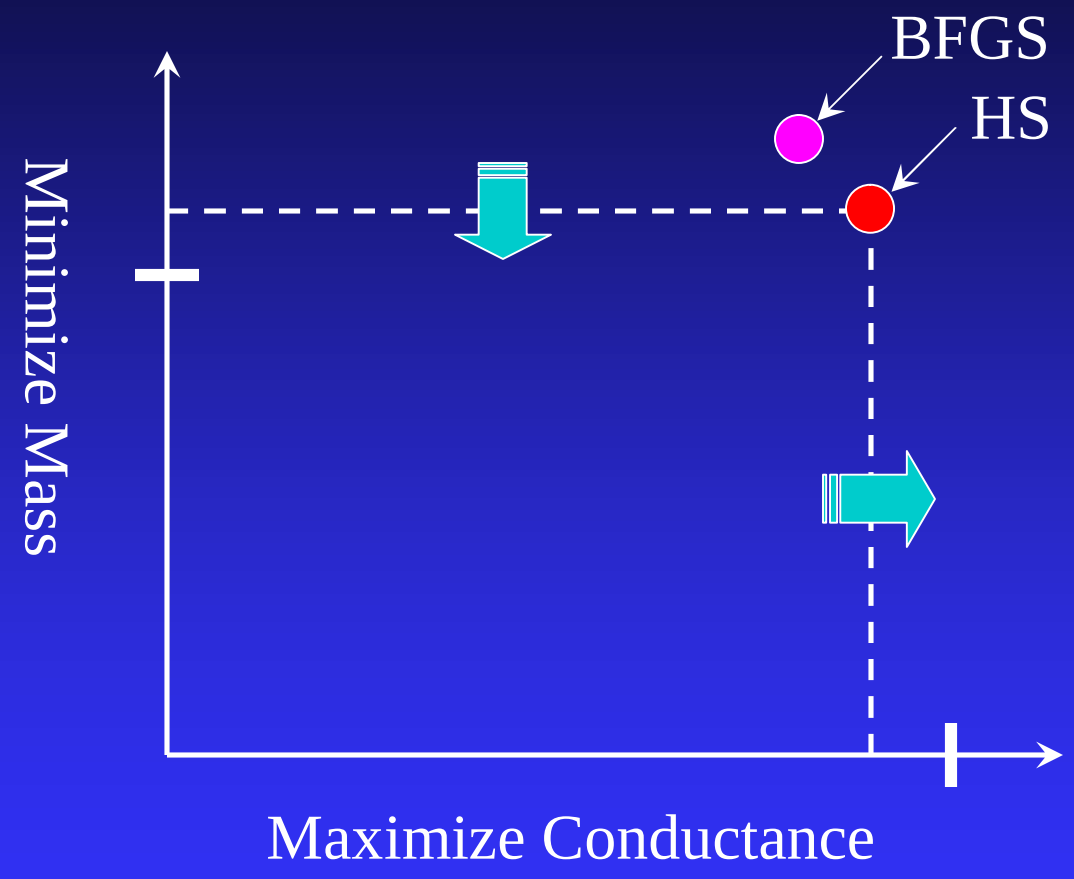
Satellite Heat Pipe Design

$$G = f(L_f, L_c, t_f, t_b, T_{op}) = 0.3745378 - 0.9352909\ tb$$
$$+ 1.01612\ tb^2 + 0.02324128\ L_c - 0.00720999\,3\,L_c^2$$
$$+ 0.00183837\,9\,L_f - 0.00005379\,707\,L_f^2 + 0.02447391\ t_f$$
$$+ 0.00230458\,3\,t_f^2 - 0.00064834\,11\,T_{op} - 0.00000092\,32971\,T_{op}^2$$
$$- 0.02259702\ t_b\ L_c - 0.00473565\,2\,t_b\ L_c^2 + 0.1102442\ t_b^2\ L_c$$
$$- 0.00970253\,3\,t_b^2 L_c^2 + 0.00538221\,1\,t_b L_f - 0.00009540\,484\,t_b L_f^2$$
$$+ 0.00515048\ t_b^2\ L_f - 0.00012325\,24\,t_b^2\ L_f^2 + 0.2972589\ t_b\ t_f$$
$$- 0.1052935\ t_b\ t_f^2 - 0.5422262\ t_b^2\ t_f - 0.1829687\ t_b^2\ t_f^2$$

$$M = f(L_f, L_c, t_f, t_b) = (1313.877 - 75.5\,L_c + 11.0\,L_c^2$$
$$+ 1.402597\,L_f - 1.278314E\text{-}15\,L_f^2 + 62.38776\,t_f$$
$$- 6.122449\,t_f^2 - 380.8\,t_b + 1120\,t_b^2) \times 21$$

$$Z = f(L_f, L_c, t_f, t_b, T_{op}) = \left|\frac{G - G^*}{G^*}\right| + \left|\frac{M - M^*}{M^*}\right|$$

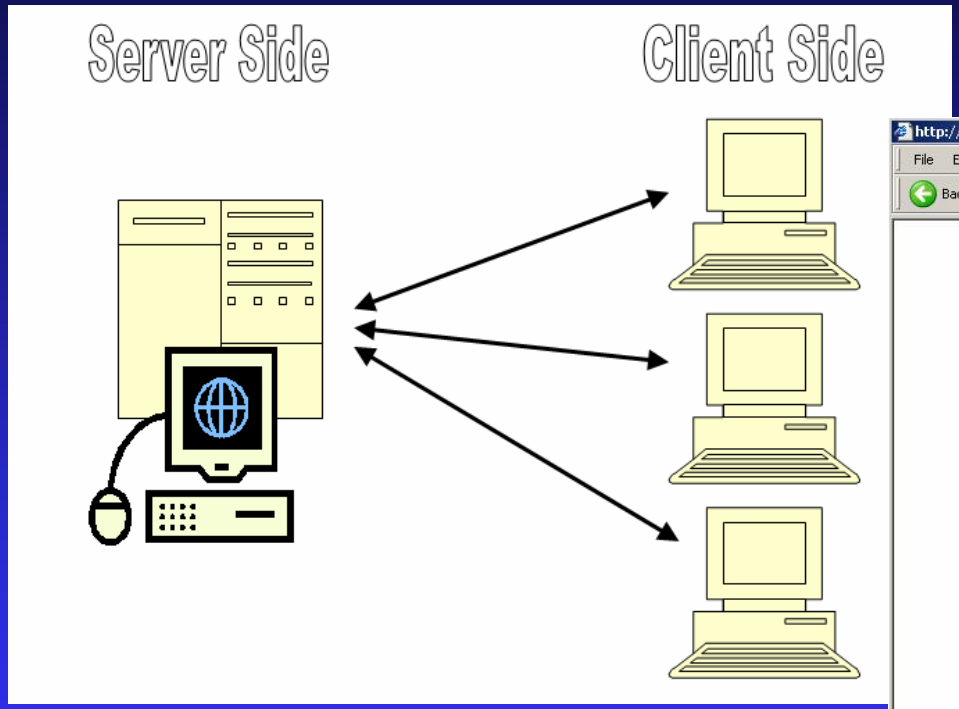
Satellite Heat Pipe Design

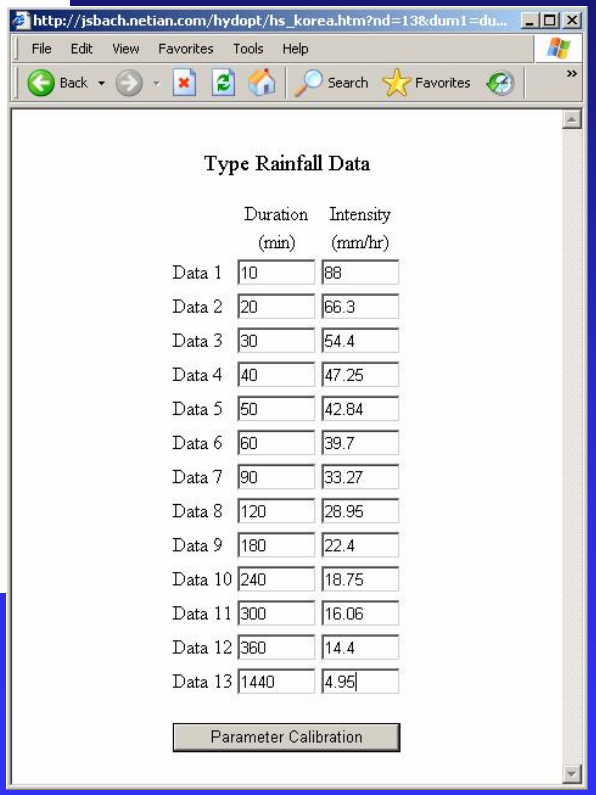


BFGS: Mass =25.9 kg, Conductance = 0.3808 W/K

HS: Mass = 25.8 kg, Conductance = 0.3945 W/K

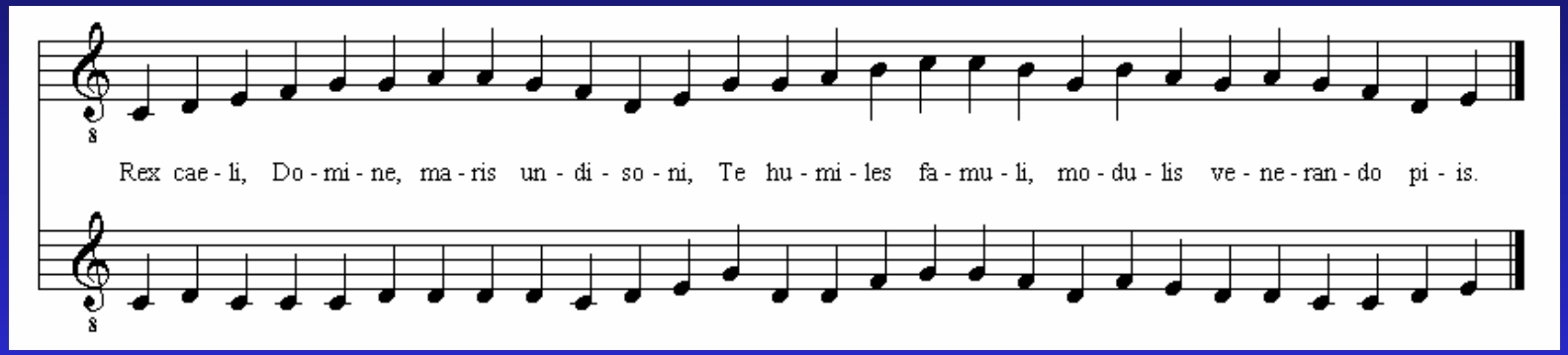
Web-Based Parameter Calibration





RMSE: 1.305 (Powell), 0.969 (GA), 0.948 (HS)

Music Composition – Medieval Organum



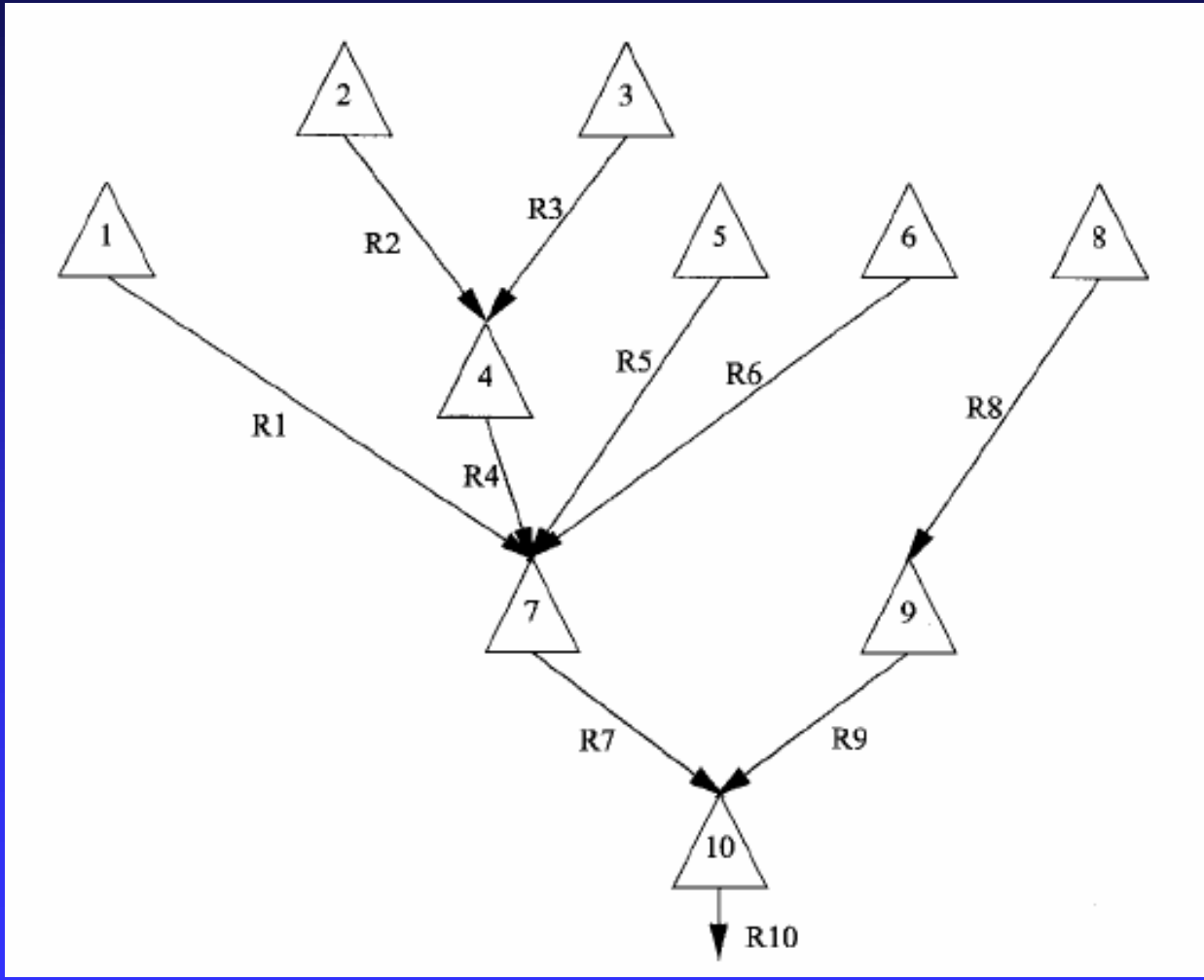
Interval	Rank	Interval	Rank
Fourth	1	Fifth	2
Unison	3	Octave	3
Third	4	Sixth	4
Second	5	Seventh	5

Sudoku Puzzle

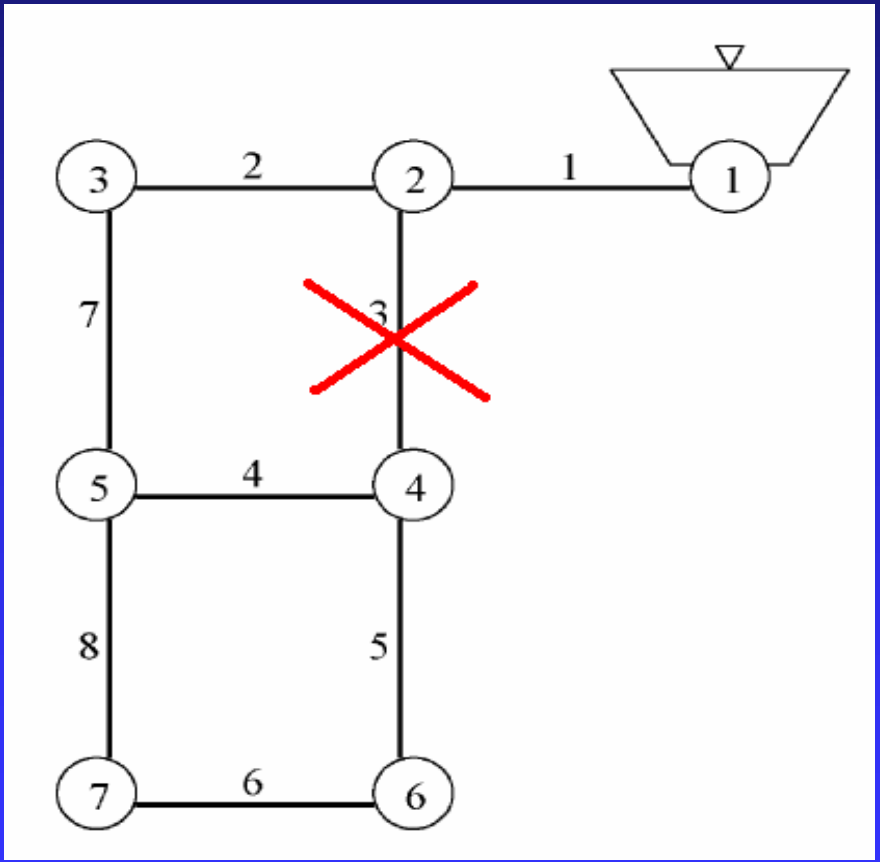
2	9	4	5	3	8	7	6	1
5	6	1	2	7	9	3	4	8
8	3	7	1	6	4	2	5	9
7	4	9	8	1	3	6	2	5
6	2	3	9	4	5	1	8	7
1	8	5	7	2	6	9	3	4
4	5	2	6	9	1	8	7	3
3	1	6	4	8	7	5	9	2
9	7	8	3	5	2	4	1	6

Future Study of Harmony Search

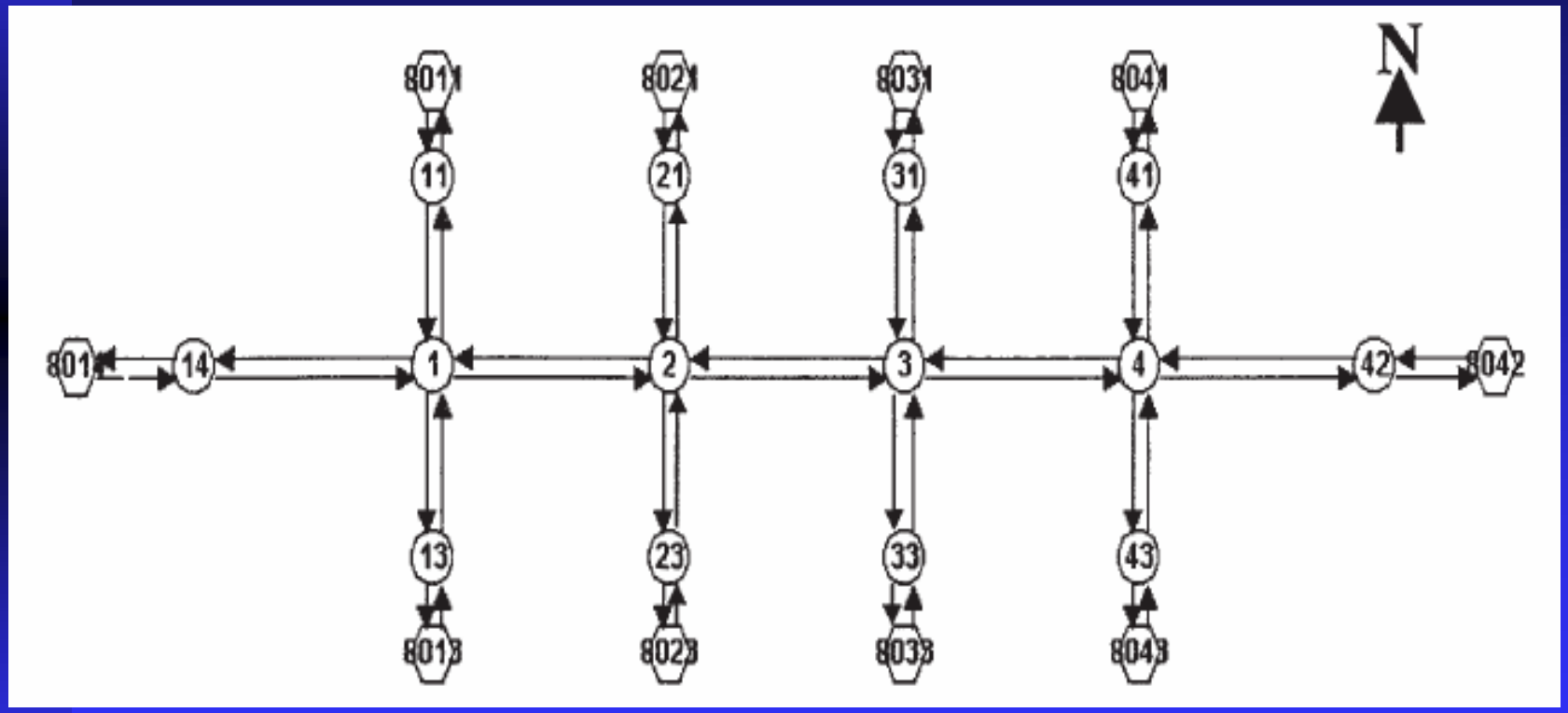
Multiple Reservoir Operation



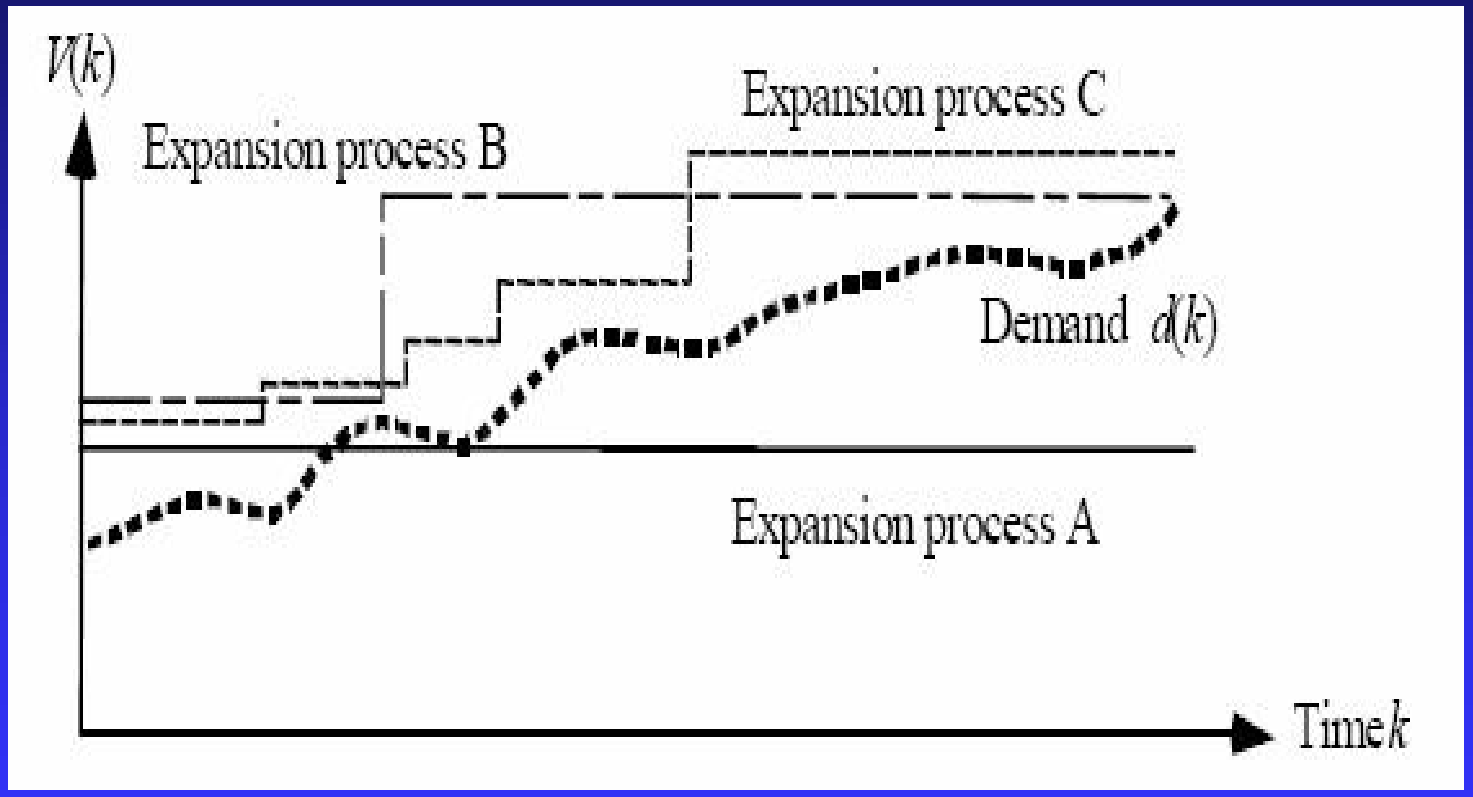
Water Network Design with Pipe Failure



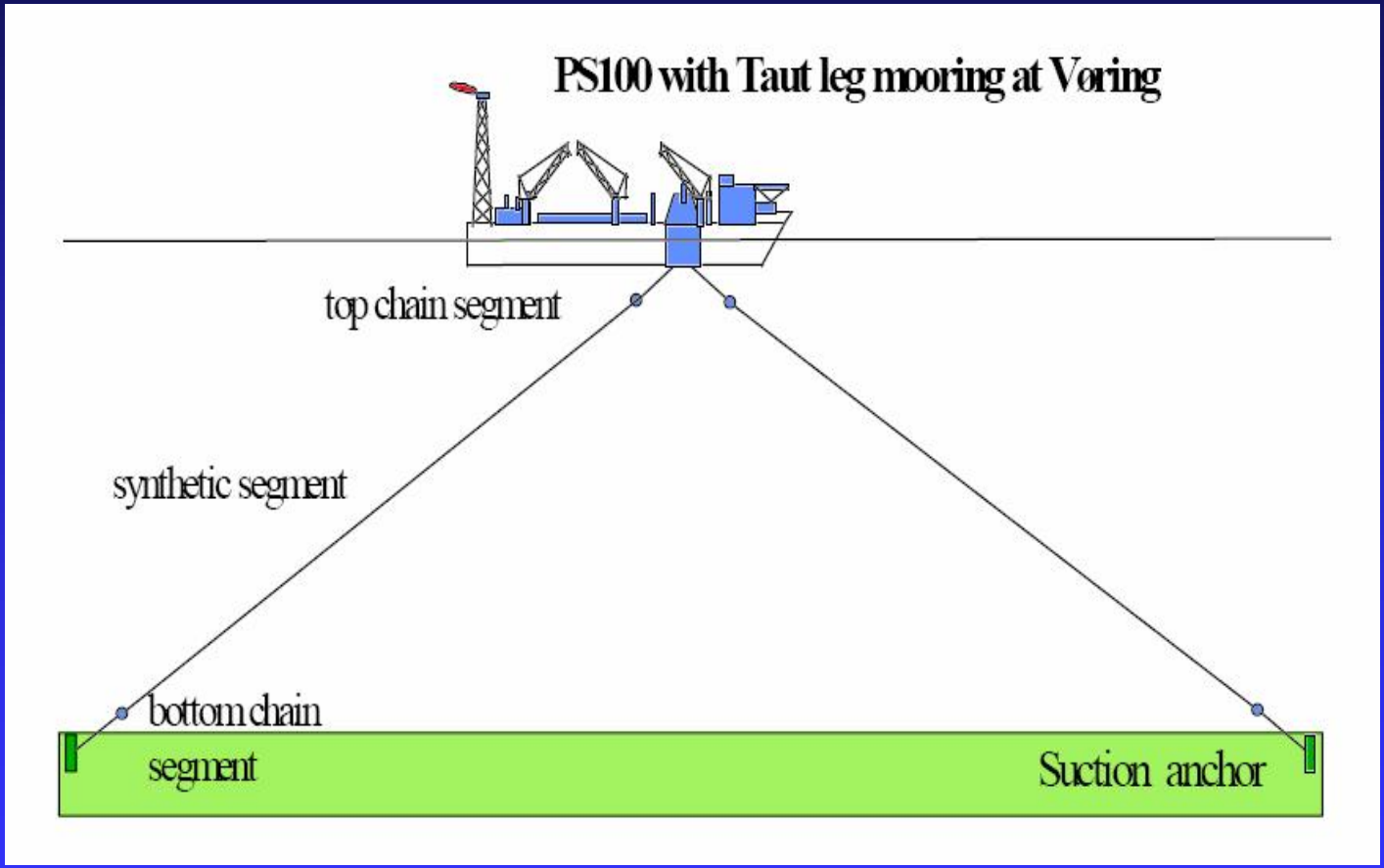
Traffic Signal Operation



Economic Power Plant Planning



Oceanic Oil Structure Design



Bioinformatics: Multiple Sequence Alignment

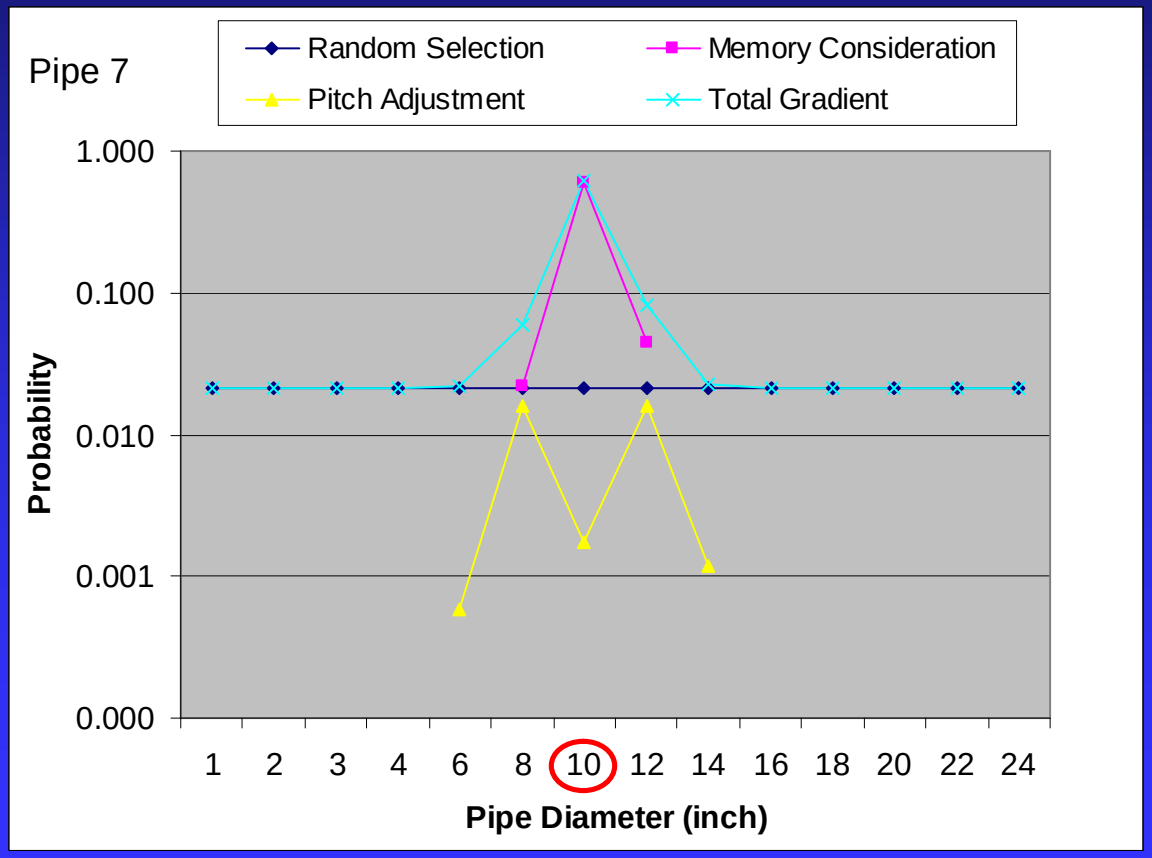
```
                    *    * * ***  *          *****            * *** ****   ***
 1  P.flavescens-N   GUU-GGGAGGAAGGGCA-GUAA-GCGAAUAC-CUUG-CUGUUUUGACGUUACCGACAGAA
 2  P.straminea-IA   GUUUGGGAAGAAGGGUU-UUAG-AUUAAUAC-UCUG-CAAUUUUGACGUUACCGACAGAA
 3  P.mucidolens-I   GUU-UGGAGGAAGGGUU-GUAG-AUUAAUAC-UCUG-CAAUUUUGACGUUACCGCCAGAA
 4  P.synxantha-IA   GUU-GGGAGGAAGGGUU-GUAG-AUUAAUAC-UCUG-CAAUUUUUACGUUACCGGCAGAA
 5  P.azotoformans   GUU-GGGAGGAAGGGUU-GUAG-AUUAAUAC-UCUG-CAAUUUUGACGUUACCGACAGAA
 6  P.fluorescens-   GUU-GGGAGGAAGGGCA-UUAA-CCUAAUAC-GUUA-GUGUUUUGACGUUACCGACAGAA
 7  P.aureofaciens   GUU-GGGAGGAAGGGUACUUA--CCUAAUAC-GUGA-GUAUUUUGACGUUACCGACAGAA
 8  P.chlororaphis   GUU-GGGAGGAAGGGUACUUA--CCUAAUAC-GUGA-GUAUUUUGACGUUACCGACAGAA
 9  P.taetrolens-I   GUU-GGGAGGAAGGGCAUUUAA-CCUAAUAC-GUUA-GUGUUUUGACGUUACCGCCAGA-
10  P.syringae-ATC   GUU-GGGAGGAAGGGCAGUUA--CCUAAUAC-GUAA-CUGUUUUGACGUUACCGACAGAA
11  P.caricapapaya   GUU-GGGAGGAAGGGCAGUUA--CCUAAUAC-GUGA-CUGUUUUGACGUUACCGACAGAA
12  P.amygdali-ATC   GUU-GGGAGGAAGGGCAGUUA--CCUAAUACCGUGA-UUGUUUUAACGUUACCCACAGAA
13  P.agarici-ATCC   GUU-GGGAGGAAGGGCA-UUAA-CCUAAUAC-GUUA-GUGUUUUGACGUUACCGACAGAA
14  P.fulva-IAM_15   GUU-GGGAGGAAGGGUU-GUAG-AUUAAUAC-UCUG-CAAUUUUGACGUUACCGACAGAA
15  P.putida-IAM_1   GUU-GGGAGGAAGGGCA-UUAA-CCUAAUAC-GUUA-GUGUUUUCACGUUACCGACAGAA
16  P.mendocina-AT   GUU-GGGAGGAAGRGCAUGUAACGYUAAUACCGUURCGUGUUUUGACGUUACCRACAGAA
17  P.alcaligenes-   GUU-GGGAGGAAGGGCA-GUAA-GUUAAUAC-CUUG-CUGUUUUGACGUUACCAACAGAA
18  P.oleovorans-I   GUU-GGGAGGAAGGGCUCAUA--GCGAAUAC-CUGU-GAGUUUUGACGUUACCAACAGAA
19  P.luteola-IAM_   GCU-GGGAGGAAGGGUUGUAA--CCUAAUAC-GUUG-CAGCUUUGACGUUACCAGCAGAA
```

Why HS outperforms MP?

- Easier Computations
  - No Complicate Calculus (Density Derivative)
  - No Initial Assumption
- Avoiding Local Optima
  - Multi-Solution Vectors
  - Random Consideration (HMCR)
- Efficient Core-Memory Usage
  - No-Enumeration

Density Derivative

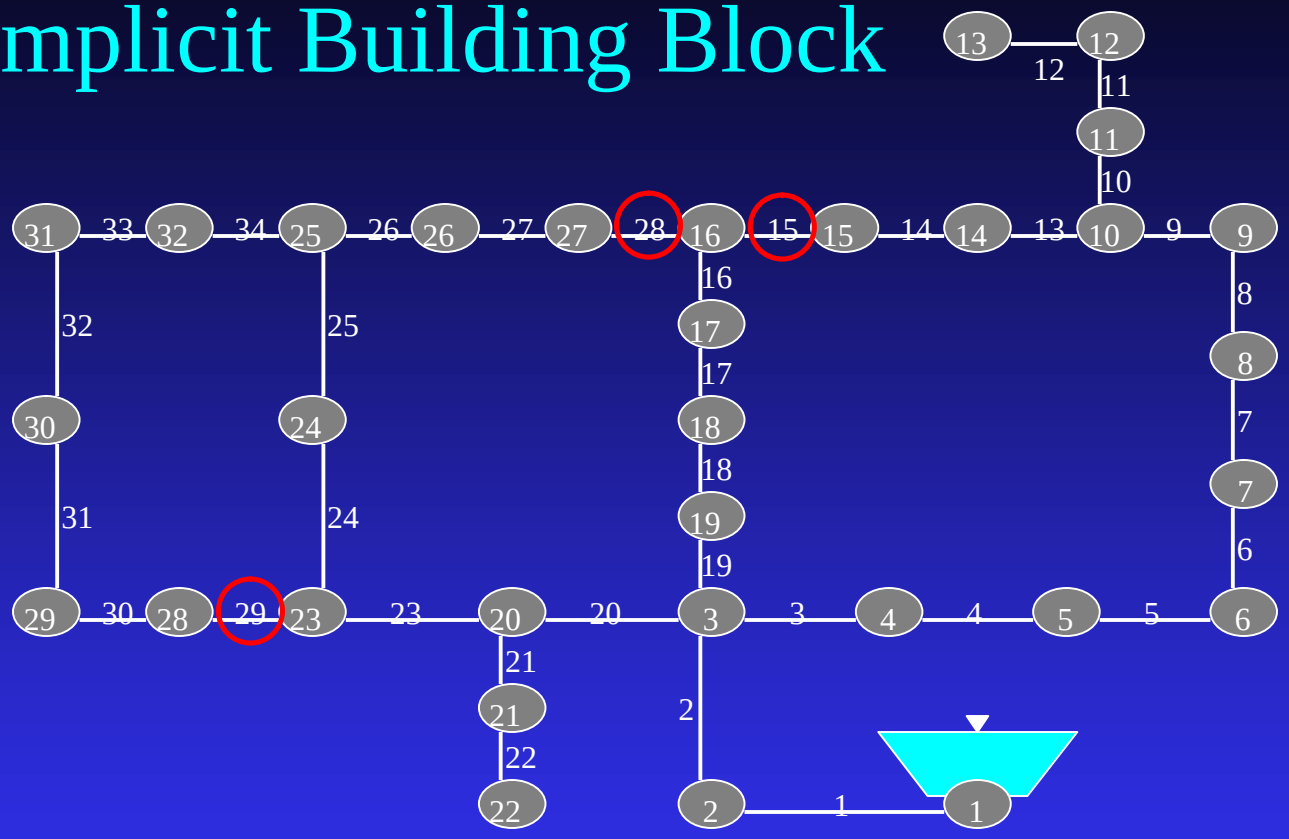
$$\frac{\partial f}{\partial x_i} = \frac{1}{K_i} \cdot (1 - HMCR) + \frac{n(x_i(k))}{HMS} \cdot HMCR \cdot (1 - PAR) + \frac{n\left(x_i(k-m)\right)}{HMS} \cdot HMCR \cdot PAR$$



Why HS outperforms GA?

- Multi-Vector Consideration
  - All Vectors in Harmony Memory
- Faster Computation
  - One Iteration in HS is Faster than that in GA
  - Binary $\leftrightarrow$ Decimal (Grey) Conversion
- Accurate Precision
  - Real Number Coding
- Correlation among Variables
  - Location in Chromosome Map (Building Block)

Implicit Building Block



Explicit Building Block

$$x_i \leftarrow f\left(x_j\right), \underset{j}{Max}\left\{Corr\left(x_i, x_j\right)\right\}$$

Question for Harmony Search?

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